

Software Engineering (Sessional) Project Proposal

Course: CSE-356

Sportzfy

On-Demand Sports Turf Booking Platform

Team Members

Mohammad Sakib Hasan

ID: 2204051

Ayan Barua

ID: 2204053

Mahmud Hasan

ID: 2204040

Department of Computer Science and Engineering (CSE)
Chittagong University of Engineering & Technology (CUET)
Chattogram-4349, Bangladesh

1. Introduction / Background

In Bangladesh, five-a-side and seven-a-side football turfs have become the default venue for casual and competitive matches among students, office groups, and local clubs, especially in cities like Chattogram and Dhaka. Despite this growing demand, the booking process remains almost entirely manual: players call the turf owner directly, message on Facebook pages, or simply show up and hope a slot is free. This leads to frequent double-bookings, no centralized way to compare turfs by price, location, or reviews, and no visibility into real-time slot availability.

Bangladesh already has strong precedents for solving exactly this kind of on-demand coordination problem digitally. Foodpanda turned scattered restaurant ordering into a single real-time marketplace, Pathao did the same for ride-hailing and logistics, and bdtickets centralized bus and event ticketing with instant, guaranteed seat confirmation. Sportzfy applies this same on-demand marketplace model to turf booking: instead of a static listing site, it behaves like a live booking engine where slot availability updates in real time, payment locks the slot instantly, and both players and turf owners get instant confirmation — removing the phone calls, guesswork, and double-booking that plague the current manual system.

2. Project Objectives

Main Objective

To design and develop Sportzfy, an on-demand turf booking platform that lets players discover nearby turfs and instantly reserve and pay for a guaranteed time slot in real time — the way Foodpanda, Pathao, and bdtickets turned food, rides, and tickets into instant, reliable, on-demand services.

Specific Objectives

- Enable customers to search turfs by location, sport type, and time, and view live (real-time) slot availability instead of static schedules.
- Provide instant slot locking and secure online payment .
- Give turf owners a self-service dashboard to list turfs, set dynamic pricing, manage slots, and track bookings and earnings without manual coordination.
- Integrate AI-driven features — smart slot recommendations, demand-based dynamic pricing suggestions, and booking-pattern predictions — to improve both discovery and turf utilization.
- Build a scalable, secure system architecture capable of handling concurrent bookings across multiple turfs without data conflicts.

3. Scope of the Project

What will be included

- Web-based platform (responsive, mobile-friendly) for three user roles: Customer, Turf Owner, and Administrator.
- Real-time turf search, filtering (location, sport, price, rating), and live slot availability.
- Instant slot booking with online payment integration and automatic booking confirmation.
- Turf owner dashboard for listing management, slot/pricing control, and booking analytics.
- AI-powered recommendation engine for turfs and optimal booking time slots.
- Review and rating system for completed bookings.
- Admin panel for turf approval, user management, and platform monitoring.

What will not be included

- Physical turf maintenance, equipment rental, or on-site staffing services.
- A native mobile application (Android/iOS) — planned as a future extension beyond this course scope.
- In-person customer support or call-center operations.

4. Proposed Features

- **Real-Time Search & Discovery:** Live turf discovery — search and filter turfs by location, sport type, price range, and availability, similar to browsing restaurants on Foodpanda.
- **Instant Slot Locking:** a persistent hold-and-release mechanism ensures a slot is locked the moment a customer starts booking and released automatically if payment is not completed in time — eliminating double-booking.
- **One-Tap Payment & Confirmation:** integrated mobile wallets (bKash, Nagad, Rocket) and card payments, with booking confirmed only after successful payment, mirroring bdtickets' guaranteed-seat model.
- **Turf Owner Dashboard:** owners add/edit turfs, set time-slot pricing, view real-time bookings, and manage availability from a single dashboard.
- **Review & Rating System:** ratings and written reviews on completed bookings, visible publicly to build trust in turf quality.
- **Admin Control Panel:** turf approval workflow, user moderation, and platform-wide analytics.

AI Integration

Sportzfy embeds AI across the discovery and pricing layers rather than treating it as an add-on:

- **Smart Slot Recommendation:** analyzes a customer's past bookings, preferred sport, and typical playing times to recommend the best-fit turf and time slot.
- **Demand Prediction & Dynamic Pricing Suggestions:** models historical booking density per turf to forecast peak-demand windows and suggest price adjustments to turf owners .

5. Methodology / Approach

Chosen SDLC Model

The project will follow the Incremental Model of the Software Development Life Cycle. Core modules — authentication, turf listing, and booking — will be delivered first as a working base, with payment integration, the AI recommendation engine, and the admin panel added in subsequent increments.

Technology Stack

- Frontend: Next.js (React), HTML, CSS, and JavaScript to deliver a responsive, SEO-optimized user interface.
- Backend: FastAPI (Python) utilizing WebSocket connections for real-time synchronization of slot availability.
- Database: PostgreSQL for robust relational storage of user data, turf listings, schedules, and transactions.
- Authentication: Secure JWT-based identity management providing role-based access for Customers, Owners, and Admins.

AI Technology

- Python (scikit-learn / TensorFlow) for the demand-prediction and recommendation models.
- Model outputs served via a lightweight API layer consumed by the Next.js frontend.

6. Expected Outcomes / Deliverables

- **Final Working Software:** a fully functional web platform where customers can search, book, and pay for turf slots in real time, and owners can manage their listings end-to-end.
- **Key AI-Powered Features:** a working smart recommendation engine and demand-based pricing insight module, demonstrably reducing manual guesswork in turf selection and pricing.
- **Documentation:** complete SDLC documentation including SRS, UML diagrams (use case, class, activity, sequence), DFDs, ER diagrams, and a system/acceptance test plan.

7. Tools & Resources Required

Hardware

- Development laptops/computers with a minimum of 8GB RAM.
- Stable internet connection for real-time database testing.
- Cloud server/hosting for live deployment and demo.

Software / APIs / Cloud Services

- Core Development: FastAPI serving as the backend framework with PostgreSQL for relational data persistence.
- Financial Integration: sandbox APIs for bKash, Nagad, and Rocket to facilitate secure payment processing.
- Workflow Tools: GitHub for version management, Figma for design prototyping, and Postman for interface validation.
- Intelligence Layer: Python-based ML libraries including scikit-learn and TensorFlow for the smart recommendation engine.

8. Conclusion

Sportzfy reimagines turf booking the way Foodpanda reimagined food ordering and Pathao reimagined rides and logistics — by replacing a fragmented, phone-call-driven process with a single, real-time, on-demand marketplace. Its core innovation lies in combining instant slot locking and guaranteed payment confirmation, in the spirit of bdtickets' seat-guarantee model, with an AI layer that actively improves discovery for customers and revenue optimization for turf owners. Beyond fulfilling the course's software engineering requirements, Sportzfy addresses a genuine, everyday coordination problem for Bangladesh's growing football and sports community, with clear potential to scale into a real, locally relevant product.